

National University of Computer and Emerging Sciences



Lab # 08

Programming Fundamentals Lab

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Semester	Fall 2023
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Instructions:

1. First think about problem statement and then write/draw your logic on paper.
2. For each task in the manual create a new C++ program with the naming convention as follows:
TASK-NO 5. **Mention what is happening in each line of code using comments.**
3. Write all codes one by one with proper numbering and also paste screen shot of each problem using snipping tool (default screen capture software in windows) on Microsoft word file.
4. Submit your file with this naming convention **ROLLNO_SECTION_GROUPNO_LABNO**
5. Submit your lab on **Google Classroom**.
6. Plagiarism is strictly prohibited. Copied assignments will get **NEGATIVE MARKS** for both the source and destination.

Objectives:

- Practice of selection (switch, ternary operator) and Repetition (types of while loops)

Problem#01: Selection (if-else if-else)

Write a C++ program for a bank account and display the following menu to the customers. Ask the user to input his/her **choice** (1-6 int value) and then perform that operation i.e. user entered his choice '3' then you will ask him again to input the amount to be deposited in the account.

```
~~~~~WELCOME~~~~~
Enter Your Choice
1. Enter name, Account number, Account type
2. Balance Enquiry
3. Deposit Money
4. Show Total balance
5. Withdraw Money
6. Cancel
3
Enter amount to be Deposited
5000
~~~~~WELCOME~~~~~
Enter Your Choice
1. Enter name, Account number, Account type
2. Balance Enquiry
3. Deposit Money
4. Show Total balance
5. Withdraw Money
6. Cancel
4
Total balance is: 84000
```

Problem#02: Selection (Switch)

You need to create a classifier using **switch** statement. Let say the name of your classifier program is “**Alpha Dict**”. The user of the Alpha Dict will always enter a character between the range of ‘A’ to ‘Z’ (Only Capitals), and then get result either it is “Vowel” OR “Consonant”. You are only allowed to use Switch statement to build logic for Vowel and Consonant, but you can use if-else to check whether the character is in the range of (A-Z). If the character is not in the expected range then there should be proper exception handling.

Problem#03: Repetition (Reverse Loop)

Write a program that prints all the odd numbers in the range [min - max] in descending order.

Note: Where ‘min’ and ‘max’ are integers variables input from the user.

Problem#04: (Ternary Operator)

You only allowed to use ternary operator for decision. Write a computer program that take three numbers from user and check whether these are equal OR not. In case of equal print message “Numbers are equal” and in case of not equal print message should be “Numbers are not equal.”

Hint: You can use multiple Ternary Operator

Problem#05: (While Loop – if)

Write a program that takes 10 numbers as inputs from user in 1 single variable. Finally, prints the **minimum** of all those numbers.

Note: You must take the inputs in a loop.

Problem#06: (Reverse Loop)

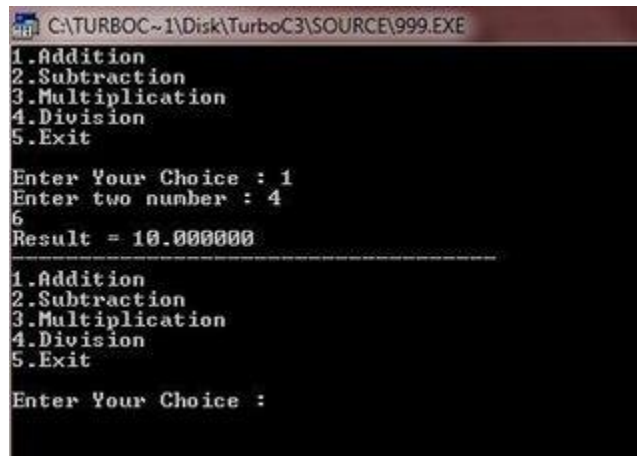
Write a program that takes two integer numbers into variables **X** and **Y** as input from the user, and calculates X^Y using loop. Library **pow** functions is strictly not allowed.

Hint: $2^5 = 2 * 2 * 2 * 2 * 2 = 32$

Problem#07: (Switch)

Write a C++ program for making simple calculator with basic arithmetic operations. First, display the following options (1-6) and then ask the user to input his/her choice 1-6 followed by two numbers.

Sample: user entered "1" that means two numbers would be added together and print the result.



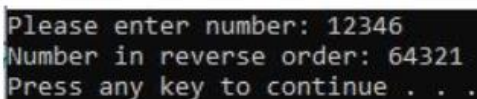
```
C:\TURBOC~1\Disk\TurboC3\SOURCE\999.EXE
1.Addition
2.Subtraction
3.Multiplication
4.Division
5.Exit

Enter Your Choice : 1
Enter two number : 4
6
Result = 10.000000
-----
1.Addition
2.Subtraction
3.Multiplication
4.Division
5.Exit

Enter Your Choice :
```

Problem#08: (Sentinel While Loop)

Write a program that input a number from user and display that number in reverse order. Your entered number must be non-zero for example:



```
Please enter number: 12346
Number in reverse order: 64321
Press any key to continue . . .
```